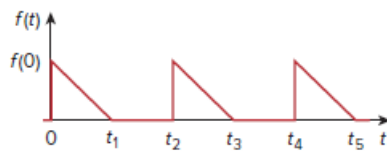


Problem

Using Fig. 17.51, design a problem to help other students better understand how to determine the exponential Fourier series from a periodic wave shape.

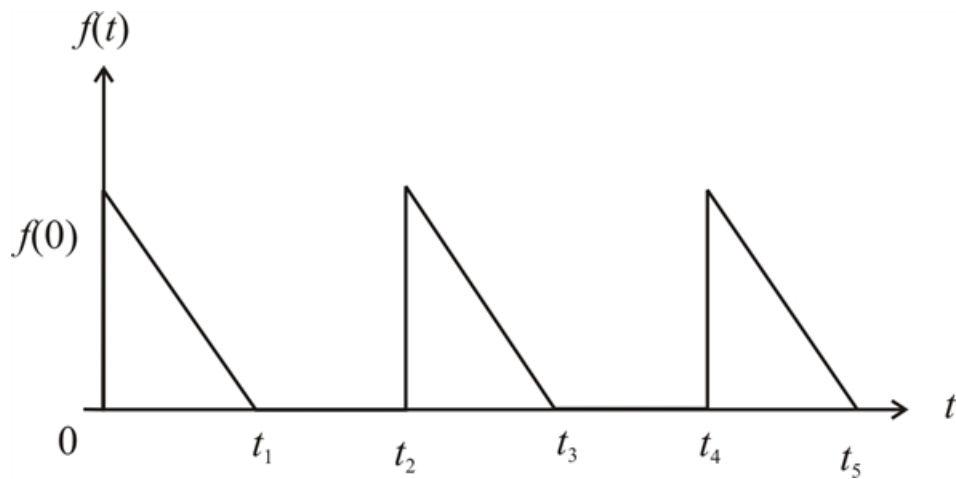
Figure 17.51:



Step-by-step solution

Step 1 of 5

Given periodic wave shape is,



Step 2 of 5

The function is

$$f(t) = \begin{cases} f(0) - \frac{f(0)}{t_1}t & 0 < t < t_1 \\ 0 & t_1 < t < t_2 \end{cases}$$

Since $T = t_2$

$$\omega_0 = \frac{2\pi}{T}$$

$$\omega_0 = \frac{2\pi}{t_2}$$

Step 3 of 5

The coefficients of exponential Fourier series are,

$$C_0 = \frac{1}{T} \int_0^T f(t) dt$$

$$C_0 = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} \left(f(t) - \frac{f(t_1)(t - t_1)}{t_2 - t_1} \right) dt$$

$$C_0 = \frac{f(t_1)}{t_2 - t_1} \left[t_1 \left(\frac{t - t_1}{t_2 - t_1} \right) - \left(\frac{t^2 - t_1^2}{2(t_2 - t_1)} \right) \right]$$

$$C_0 = \frac{f(t_1)}{t_2 - t_1} \left[t_1^2 - \frac{t_1^2}{2} \right]$$

$$C_0 = \frac{f(t_1)}{t_2 - t_1} \left[\frac{t_1^2}{2} \right]$$

$$\therefore C_0 = \frac{f(t_1)}{2} \left[\frac{t_1}{t_2 - t_1} \right]$$

Step 4 of 5

$$C_n = \frac{1}{T} \int_0^T f(t) e^{-jn\omega_0 t} dt$$

$$C_n = \frac{1}{t_2 - t_1} \int_{t_1}^{t_2} \frac{f(t_1)(t_1 - t)}{t_2 - t_1} e^{-jn\omega_0 t} dt$$

$$C_n = \frac{f(t_1)}{t_2 - t_1} \int_{t_1}^{t_2} (t_1 - t) e^{-jn\omega_0 t} dt$$

$$C_n = \frac{f(t_1)}{t_2 - t_1} \left[\int_{t_1}^{t_2} t_1 e^{-jn\omega_0 t} dt - \int_{t_1}^{t_2} t e^{-jn\omega_0 t} dt \right]$$

$$C_n = \frac{f(t_1)}{t_2 - t_1} \left[t_1 \int_{t_1}^{t_2} e^{-jn\omega_0 t} dt - \int_{t_1}^{t_2} t e^{-jn\omega_0 t} dt \right]$$

$$C_n = \frac{f(t_1)}{t_2 - t_1} \left[t_1 \int_{t_1}^{t_2} e^{-j \frac{n2\pi}{T} t} dt - \int_{t_1}^{t_2} t e^{-j \frac{n2\pi}{T} t} dt \right]$$

Step 5 of 5

$$C_n = \frac{f(0)}{t_1 t_2} \left[t_1 \frac{e^{-j \frac{n 2 \pi}{t_2} t}}{-j \frac{2 n \pi}{t_2}} \right]_0^{t_1} - \left[\frac{t e^{-j \frac{n 2 \pi}{t_2} t}}{-j \frac{2 n \pi}{t_2}} \right]_0^{t_1} - \int_0^{t_1} \frac{e^{-j \frac{2 n \pi}{t_2} t}}{-j \frac{2 n \pi}{t_2}} dt \right]$$

$$C_n = \frac{f(0)}{t_1 t_2} \left[t_1 \frac{\left(e^{-j n 2 \pi \frac{t_1}{t_2}} - 1 \right)}{-j \frac{2 n \pi}{t_2}} - \left[\frac{t e^{-j \frac{n 2 \pi}{t_2} t}}{-j \frac{2 n \pi}{t_2}} - \frac{e^{-j \frac{2 n \pi}{t_2} t}}{\left(-j \frac{2 n \pi}{t_2} \right)^2} \right]_0^{t_1} \right]$$

$$C_n = \frac{f(0)}{t_1 t_2} \left[\frac{t_1 t_2 \left(e^{-j n 2 \pi \frac{t_1}{t_2}} - 1 \right)}{-2 j n \pi} - \left(\frac{t_1 t_2 e^{-j 2 n \pi \frac{t_1}{t_2}}}{-2 j n \pi} + \frac{e^{-j 2 n \pi \frac{t_1}{t_2}} - 1}{\frac{4 n^2 \pi^2}{t_2^2}} \right) \right]$$

$$C_n = \frac{f(0)}{t_1 t_2} \left[\frac{t_1 t_2 e^{-j 2 n \pi \frac{t_1}{t_2}} - t_1 t_2}{-2 j n \pi} + \frac{t_1 t_2 e^{-j 2 n \pi \frac{t_1}{t_2}}}{2 j n \pi} - \frac{e^{-j 2 n \pi \frac{t_1}{t_2}} - 1}{\frac{4 n^2 \pi^2}{t_2^2}} \right]$$

$$C_n = \frac{f(0)}{t_1 t_2} \left[\frac{-t_1 t_2 e^{-j 2 n \pi \frac{t_1}{t_2}}}{2 j n \pi} + \frac{t_1 t_2}{2 j n \pi} + \frac{t_1 t_2 e^{-j 2 n \pi \frac{t_1}{t_2}}}{2 j n \pi} - \frac{e^{-j 2 n \pi \frac{t_1}{t_2}} - 1}{\frac{4 n^2 \pi^2}{t_2^2}} \right]$$

$$C_n = \frac{f(0)}{t_1 t_2} \left[\frac{t_1 t_2}{2 j n \pi} - \frac{e^{-j 2 n \pi \frac{t_1}{t_2}} - 1}{\frac{4 n^2 \pi^2}{t_2^2}} \right]$$

$$C_n = f(0) \left[\frac{1}{2 j n \pi} - \frac{t_2}{t_1} \frac{\left(e^{-j 2 n \pi \frac{t_1}{t_2}} - 1 \right)}{4 n^2 \pi^2} \right]$$

Therefore, the exponential Fourier series is

$$f(t) = \frac{f(0)}{2} \left[\frac{t_1}{t_2} \right] + f(0) \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} \left[\frac{1}{2 j n \pi} - \frac{t_2}{t_1} \frac{\left(e^{-j 2 n \pi \frac{t_1}{t_2}} - 1 \right)}{4 n^2 \pi^2} \right] e^{j \frac{n 2 \pi}{t_2} t}$$